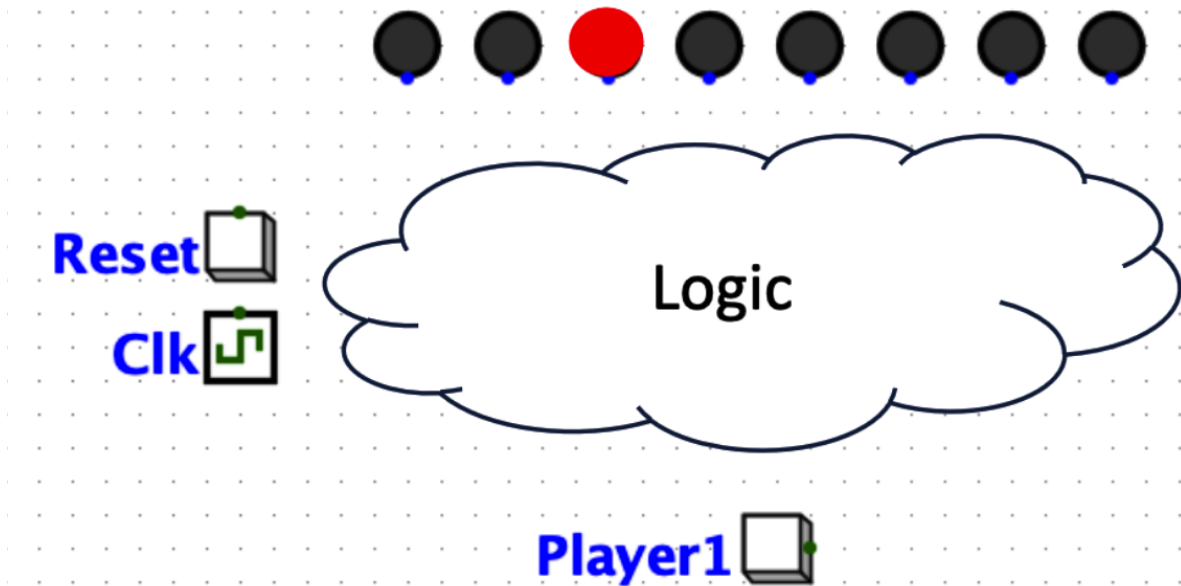


Modeling and Simulation of the Display Board in Logisim Evolution

ECE NTUA 2025-26

Digital Systems Laboratory



Due to 28/04

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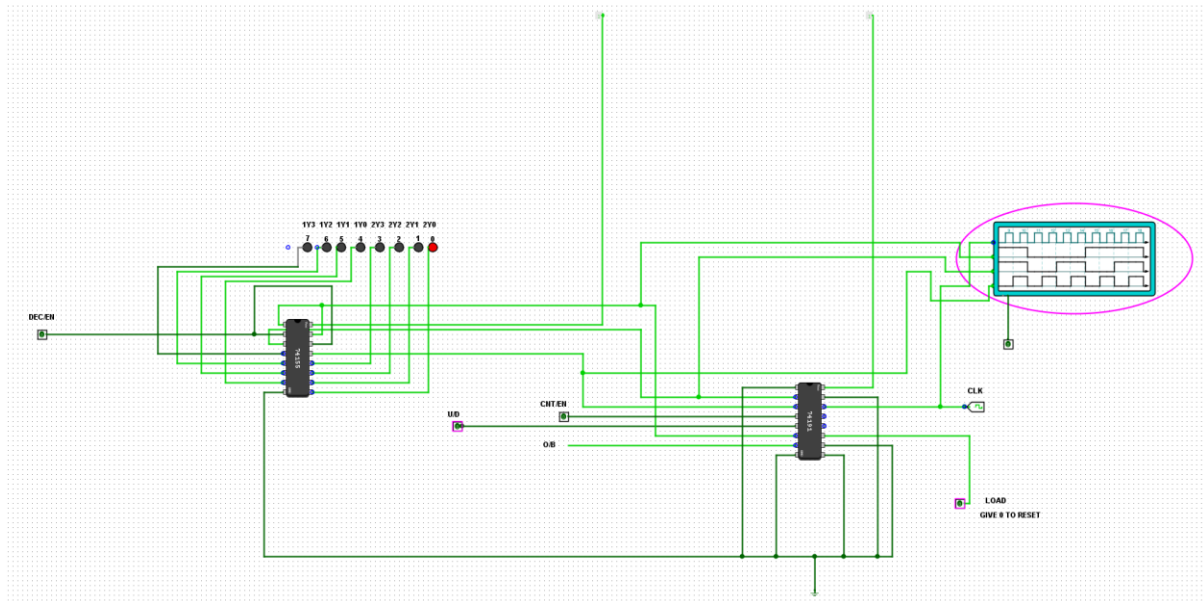
1-Purpose of the Laboratory Exercise

The primary objective of this laboratory exercise is the modeling and simulation of the 'Display' subsystem for the Pong digital game using the Logisim Evolution software tool. Following the physical implementation on a perfboard during Laboratory #3, this stage focuses on recreating the hardware's logic core in a virtual environment to validate our design and set a simulator base-backup before further integration. In the previous stage, the display's hardware part was implemented and soldered on a perfboard. From now on further control logic will be integrated to our circuit, so in order to prepare for it we simulate the display board in software in order to set the base for future control logic additions. Later on these will be physically implemented, so this simulation serves as a critical bridge toward the development of the fully autonomous game logic.

Summarizing previous sessions, our system utilizes the 74191 4-bit up/down counter, which receives clock input from a 555 timer and passes 3-bit number sequences to the 74155 decoder. The decoder turns on one led at a time based on the sequenced received from the counter. As a result, we manage to represent the ball's position across a row of 8 LEDs, ensuring that internal binary data is correctly transformed into a visible pattern.

2-Circuit Modeling in Logisim Evolution

Our Logisim software simulation of the full Display part was implemented directly reflecting the physical interfaces defined in our Lab 3 hardware setup. We aimed at transforming our handwritten notes into a software-logisim circuit, which means topography and placements resemble our handwritten sketches. Creating a high level software edition was not our goal, as this simulation needs to play the role of our Lab's real circuit simulator - "hardware debugger". At the same time, we aimed at adding a few futures that Logisim provides in order to utilize the extra capabilities given by the Software (for example digital oscilloscope, switches etc).



*Since the specific TTL components are not natively available in the Logisim library, custom Java files -found in Helios- were used to accurately simulate the 74191 and 74155 ICs.

Connections and interfaces in simulation were done as described in Lab 3 Report, with some simplifications being made for example in LEDs connections to power and ground, due to Logisim being a boolean-based simulator and not an electrical one. Our first task was to keep the main blocks safely inter-connected: clock- counter-decoder-LEDs. Outside signals/interfaces were connected to manual pins for the user to handle and observe with added text for basic understanding.

3-Control Strategy

Because the final autonomous Control Logic of the Pong game is not yet implemented at this stage, a manual control strategy was formulated to validate the Display subsystem. Mirroring the physical testing methodology we established in Lab 3, we use Logisim's simulation tools to force the external control signals to specific logic levels.

To establish normal operation = up count (for example), meaning LEDs turning on consecutively from 2y0 to 1y3 (0 to 7) we manually bind the following:

- Both enables set to 0 (active on 0)
- Up/down set to 0 (up count)
- Load pressed one time (=0) to initialize 000 = LED0 as the starting point and then binded 1 to let the process happen.

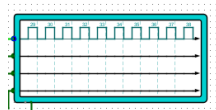
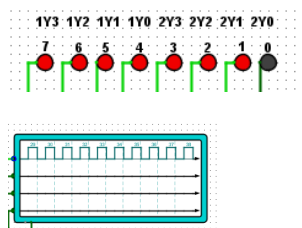
This manual state-toggling within the simulation allows us to verify the integrity of the datapath and ensure the display correctly translates the binary count into a single, shifting illuminated position at every clock cycle.

4-Interesting Simulation Scenarios and Waveforms

To thoroughly document our control strategy and prove the functional correctness of our Logisim model, we captured waveforms using the built-in oscilloscope for three critical game scenarios:

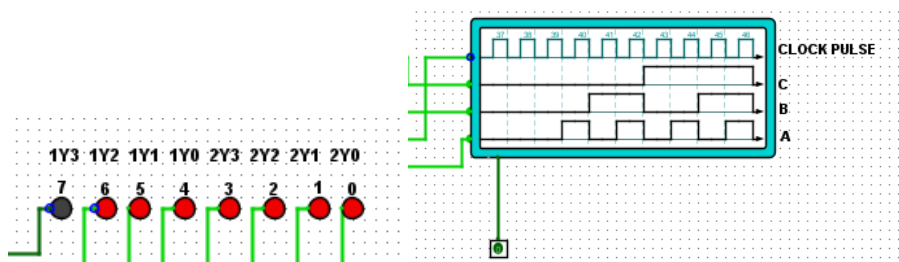
A. System Initialization

The game must start from a predictable state. In this scenario, the Reset switch is toggled to logic 0. The waveform below demonstrates that as long as the reset is held low, the 74191 counter is forced to load 0000 (due to the grounded data inputs). Consequently, the 74155 decoder steadily activates only the first LED (position 0), regardless of incoming clock pulses.



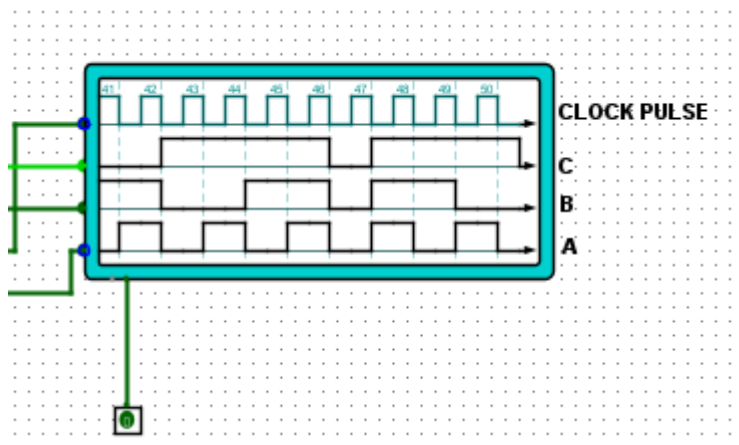
B. Sequential Movement

Once initialization is complete and Reset returns to logic 1, the system enters its standard sequential state. For this test, UpDown is set to 0. The captured waveform illustrates the clock's square wave alongside the shifting states of the counter's 3 bits (A, B, C). It clearly shows the active output shifting systematically from position 0 up to position 7, representing the ball moving across the screen.



C. Direction Reversal

A core mechanic of Pong is the ball changing direction upon hitting a paddle or a wall. To simulate this, the UpDown input is flipped from logic 0 to logic 1 during operation. The waveform captures the exact moment of this transition: the counter immediately stops incrementing and begins to decrement with each subsequent clock pulse, successfully reversing the visual sequence of the LEDs.



5-Conclusion and Implementation Notes

The transition from a physical perfboard implementation to a virtual simulation environment underscored the critical importance of precise pin mapping and thorough component understanding. Analyzing the custom Java files provided for the TTL components was essential, particularly to decode the internal logic of the 74155 demultiplexer and its specific dual-activation conditions (such as the 1G/1C and 2G/2C pins).

Furthermore, utilizing the built-in Logisim oscilloscope proved to be an invaluable tool for real-time debugging. It allowed us to visually confirm the exact synchronization between the external clock pulses, the counter's state changes, and the resulting LED transitions. Ultimately, this simulation phase successfully validates the logical robustness of our Display subsystem. It establishes a verified, working virtual foundation before we proceed to design and integrate the final, autonomous Control Logic in the upcoming laboratory sessions.